Lecture 9: Evaluating Policy Counterfactuals

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Evaluating policy counterfactuals

Remember Lecture 4: How does **systematic policy** shape the economy?

E.g., what if the Fed had followed a simple Taylor rule post-2021?

- Traditional approaches:
  1. *Structural*: build complete general equilibrium model, then change policy rule
     Smets-Wouters (2007) is the classic, Bocola-Dovis-Jørgensen-Kirpalani (2024) more recently.
     - What moments should I be targeting in estimating my model?
     - What shocks drive the cycle? Are they "dubiously structural"? Is that a problem?
  2. *Empirical*: use evidence on policy shock transmission to change policy through shocks
     Lecture Note 4: We saw Sims-Zha (1998, 2006) + modern sufficient statistics results.
- **Today**: practical applications to policy counterfactual evaluation

This lecture

1. **Identification results** [extension]
   - Finish Lecture Note 4: **Q2** [average cycle] and **Q3** [alternative history]
   - Main **sufficient statistics** result: reduce policy evaluation to ...
     a) reduced-form **second moments** [variances, covariances, autocovariances, ...]
     b) **policy causal effects**
2. **Implementation**
   - *Empirics* alone are already enough to give **a)** and at least parts of **b)**
   - Residual role of *model structure* is extrapolation: supplement empirics to get rest of **b)**
3. **Applications**
   Will consider both purely empirical as well as structural exercises, both using our sufficient statistics.

Outline

1. Extended Identification Results

2. Implementation
   - Do We Need All of $\Theta_{x,z}$?
   - Extrapolation

3. Applications
   - Empirics-Only
   - Empirics + Model Extrapolation

4. Summary

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Policy counterfactual questions

- Remember Lecture Note 4: we defined three kinds of **policy counterfactual questions**

  Q1 **Shock propagation**: how is the propagation of a particular shock affected?
  E.g., how would oil shocks propagate in the absence of a central bank reaction?

  Q2 **Business-cycle statistics**: how different would the average business cycle look?
  E.g., how would strict adherence to a Taylor rule have shaped output volatility post-WWII?

  Q3 **Alternative history**: how would a particular historical episode have unfolded?
  E.g., how would the post-covid inflation have unfolded with earlier rate hikes?

- We argued that, for **Q1**, **policy causal effects** + IRFs of the **original shock** suffice

- Next few slides: for **Q2** & **Q3**, the sufficient statistics are even easier, need ...

  a) reduced-form **second moments** [variances, covariances, autocovariances, ...]

  b) **policy causal effects**

Extension to other counterfactuals

- Can we extend our earlier logic for **Q1** to **Q2** & **Q3**?
  - Suppose that economy is subject to full vector of (policy + non-policy) shocks e_t, giving

$$y_t = \sum_{\ell=0}^{\infty} \Theta_\ell e_{t-\ell}, \quad \text{Var}(e_t) = I \tag{1}$$

    where the y_t's are observable macroeconomic outcomes, $y_t = (x_t', z_t')'$

    For each shock, the IRFs Θ_ℓ come from a linear perfect-foresight system like the one in Lecture 4.
  - Contemplate changing the policy rule to $\tilde{A}_x \boldsymbol{x} + \tilde{A}_z \boldsymbol{z} = \boldsymbol{0}$.

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    Q2 **Business-cycle statistics**: cnfct'l IRFs $\tilde{\Theta}_\ell$ give cnfct'l autocovariance function

$$\tilde{\Gamma}_y(\ell) = \sum_{m=0}^{\infty} \tilde{\Theta}_{m+\ell} \tilde{\Theta}_m'$$

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    Q3 **Alternative history**: cnfct'l outcomes at a particular date t would be [ignoring initial cond's]

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  - Immediate if we can observe e_t, just proceed shock-by-shock. But can we relax that?

Extension to other counterfactuals

Proposition

Suppose that the SVMA(∞) process (1) *is* ***invertible****. Then a)* y_t *and its autocovariance f'n* $\Gamma_y(\ell)$ *and b) policy effects* Θ_ν *suffice to construct the second and third counterfactuals.*

- Key idea: **invertibility** substitutes for knowledge of the individual shocks e_t. Why?
  - Using a), write the Wold representation as $y_t = \sum_{\ell=0}^{\infty} \Psi_\ell u_{t-\ell}$, with u_t orthonormalized.
  - Using b), apply McKay-Wolf (2023) to the Wold innovation IRFs to get the cnfctl's:

$$\sum_{\ell=0}^{\infty} \tilde{\Psi}_\ell u_{t-\ell} = \sum_{\ell=0}^{\infty} \tilde{\Theta}_\ell Q' Q e_{t-\ell} = \sum_{\ell=0}^{\infty} \tilde{\Theta}_\ell e_{t-\ell} = \tilde{y}_t$$

    where we use that $u_t = Q e_t$ and $\tilde{\Psi}_\ell = \tilde{\Theta}_\ell Q'$, by invertibility. [details in Lecture Note 5]

- Intuition: predictions based on 1-1 function of e_t = predictions based on the true e_t.

Non-invertibility in practice

Invertibility is a strong assumption. What happens when it is violated?

Visual illustration using Smets-Wouters model as DGP. Implement proposition with different info sets.

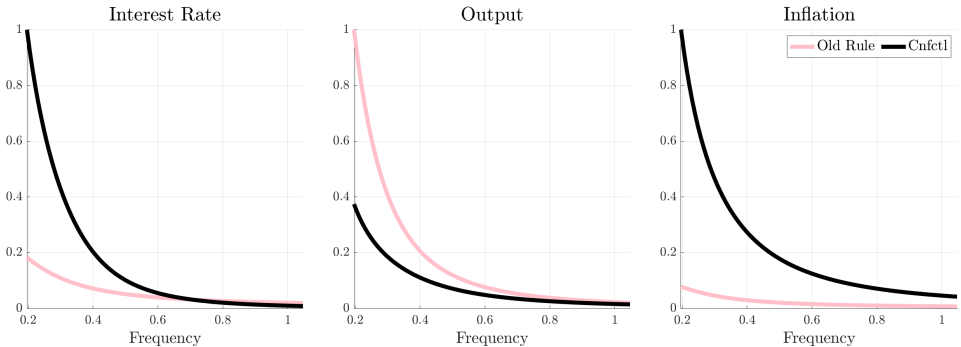


Figure shows baseline and counterfactual (true & predicted) spectral densities.

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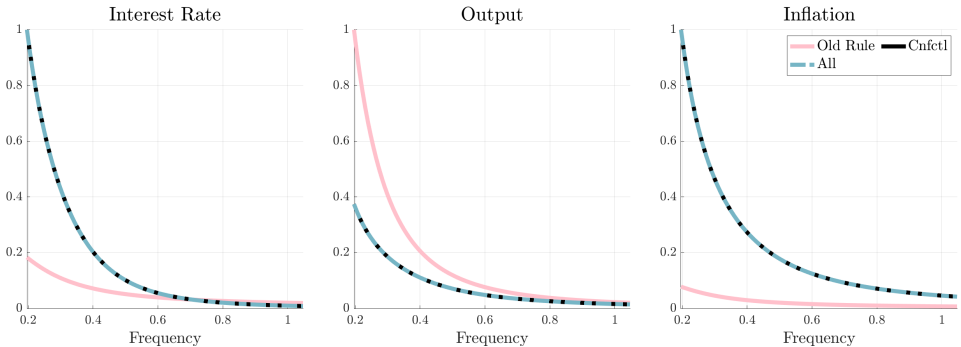


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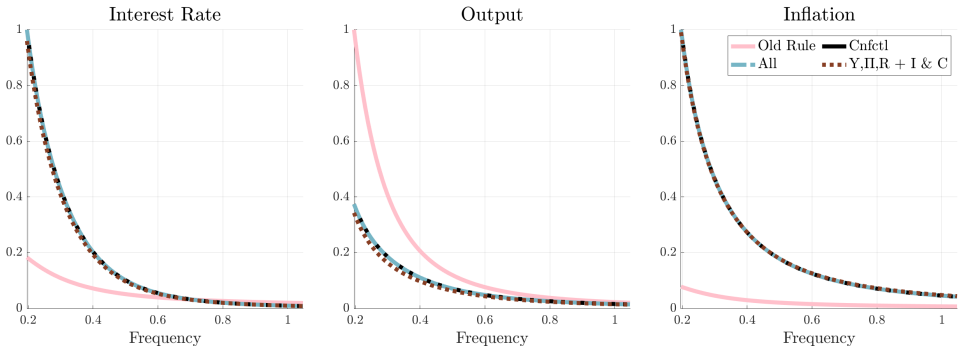


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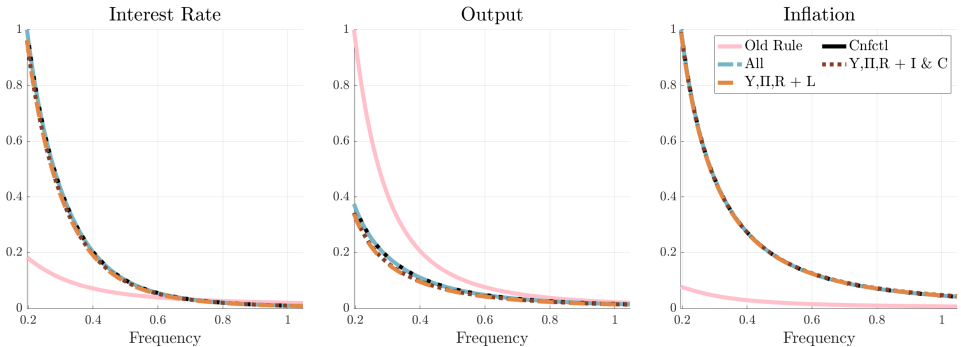


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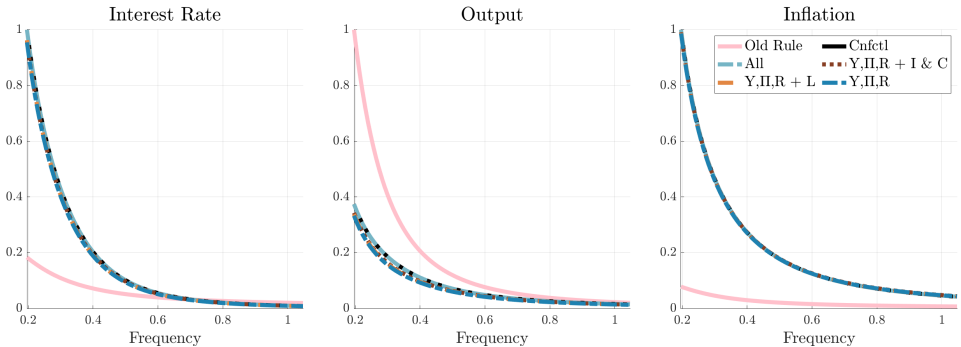
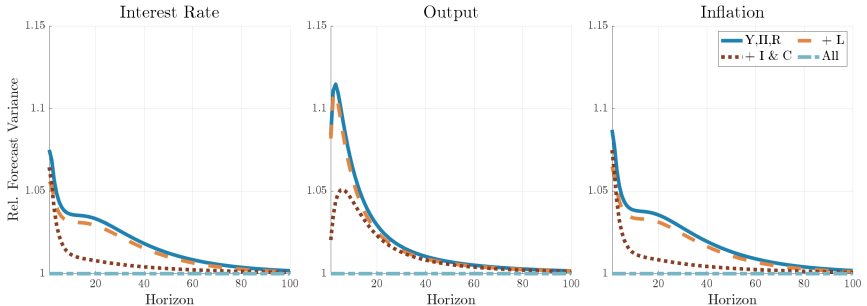


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Non-invertibility in practice

Can get close to the truth even when far from **invertibility**. Why?

Key intuition: proof works as long as fcst's based on $\{y_{t-\ell}\}_{\ell=0}^{\infty}$ = fcst's based on $\{e_{t-\ell}\}_{\ell=0}^{\infty}$.



Shows relative forecast uncertainty. Predictions were accurate because addt'l observables don't help much.

Aside: note that invertibility here plays a very different role compared to our identification discussion.

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Roadmap

How can we leverage these **identification results** in practice?

- What we need vs. what we have
  - Focus on **Q2** & **Q3**. Need: a) **autocovariance function** & b) **policy causal effects**
  - What do we have?
    a) Can get **autocovariance function** easily, using reduced-form tools
    b) We don't have all of $\Theta_{x,z}$. At best we have some parts $\theta_{x,z} \subset \Theta_{x,z}$.
- Next few slides:
  1. Could $\theta_{x,z}$ ever be enough?
  2. How can we go to $\Theta_{x,z}$?

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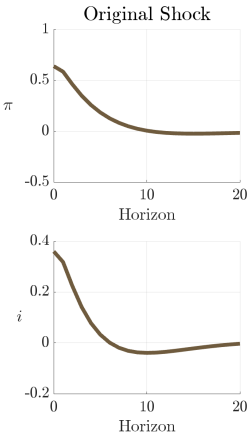
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In practice a few policy shocks *may* suffice

Q: How would this **forecast** have changed in the absence of a monetary reaction?

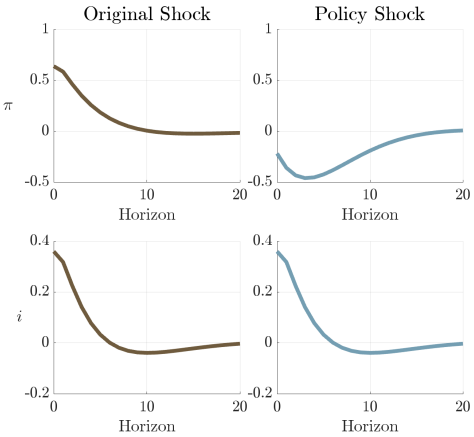
Can interpret this IRF as specific shock (for Q1), Wold innovation, or just forecast (for Q2 & Q3).



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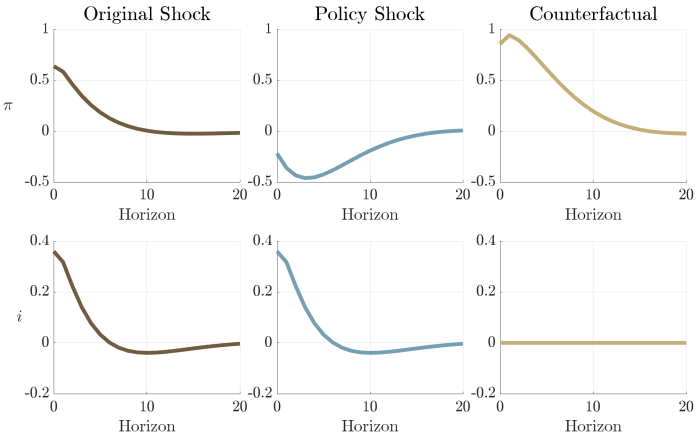


- ID result: find a **monetary shock** inducing the same rate response
  $\Rightarrow$ move $\mathbb{E}_0(i_t)$ just like the **forecast**

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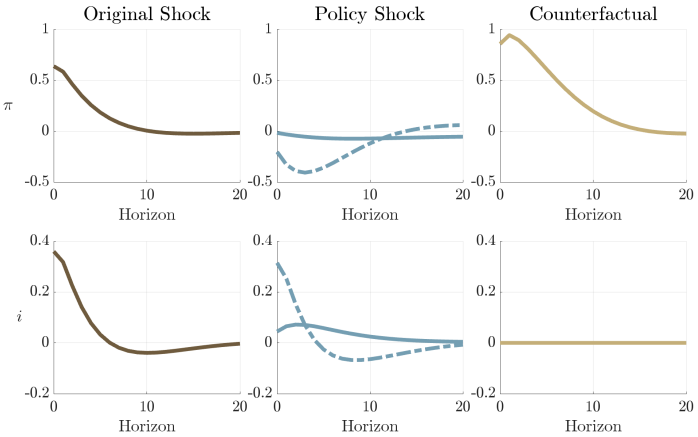


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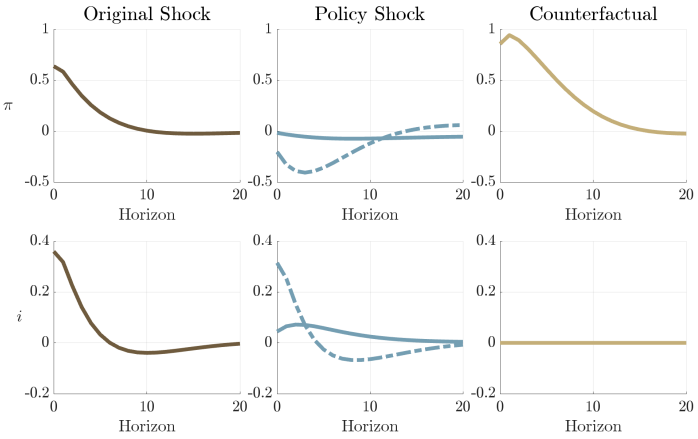


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  [Same result for combo of MP shocks.]

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- $\Longrightarrow$ if a model matches **(1)** & **(2)**, it will agree with **cnfct'l (3)**
  [Same result for combo of MP shocks.]
- Thus: for any given application, a small **# of shocks** may suffice to enforce the desired **cnfct'l rule**

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Models as extrapolation devices

What if the counterfactual of interest is not spanned by **policy shock** evidence?

- Obvious solution: use model structure to **extrapolate**
  - Suppose empirics give $\theta_{x,z} \subset \Theta_{x,z}$. Build model matching this evidence, then extrapolate.
  - How much model structure do we need for this?
    - → Need to specify $\mathcal{H}_x$ and $\mathcal{H}_z$ – the non-policy block
    - → Policy rule $\mathcal{A}_\bullet$ or non-policy shocks $\boldsymbol{\varepsilon}$ instead don't matter at all
      Sidesteps concerns about "dubiously structural" shocks. Just not needed for extrapolation.

  It's just impulse response matching! This + **autocovariance function** is all we need.
  Note however how the implementation differs from CEE (2005). No need to commit to rule!

- Implication for **modeling**: key **Q** is whether your model *changes* the extrapolation
  E.g., does HANK extrapolate differently beyond the available evidence on monetary transmission?

Re-interpreting full structural approaches

What about conventional full-blown **likelihood-based approaches**?

E.g., as in Smets-Wouters (2007) or Bocola-Dovis-Jørgensen-Kirpalani (2024).

- Through the lens of the **sufficient statistics** result:
  - **Second moments** matched [more or less] by construction, no direct discipline on **policy causal effects** [but often reported and compared with evidence]
  - So is it a problem if the shocks are "dubiously structural"? Only if:
    1. Having incorrect shocks distorts inference on **policy causal effects**
    2. We care about the nature of the shocks *per se*, e.g., for welfare reasons
- vs. just using a model for **extrapolation**: familiar efficiency vs. robustness trade-off
  - Previous slide: need to specify less, in particular can be silent on hidden shock processes
  - But: if everything is correctly specified, then full-information approach is more efficient

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Applications

- We'll now illustrate using a series of **empirical applications** to monetary policy
  1. Strategy I: only use empirical estimates of $\theta_{x,z} \subset \Theta_{x,z}$
     - What would've happened if the Fed had hiked rates earlier post-2021?
     - Post-1990, could the Fed have improved macro outcomes?
  2. Strategy II: use empirical estimates $\theta_{x,z}$ plus model extrapolation to get all of $\Theta_{x,z}$
     - Extrapolation: empirical evidence gives effects of short-lived rate changes. How do different models extrapolate?
     - Applications: flexible inflation targeting post-WWII, post-2021 w/ different eq'm selection
- Note: throughout evaluate certainty-equivalent optimal policies. No cautious behavior.
  cf. the brief discussion at the end of Lecture Note 6.

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Evaluating post-2021 rate hikes

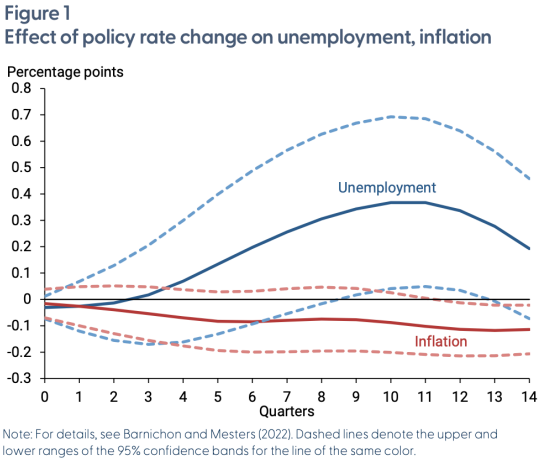


Figure shows estimated monetary policy causal effects. Source: Barnichon (2022).

Evaluating post-2021 rate hikes

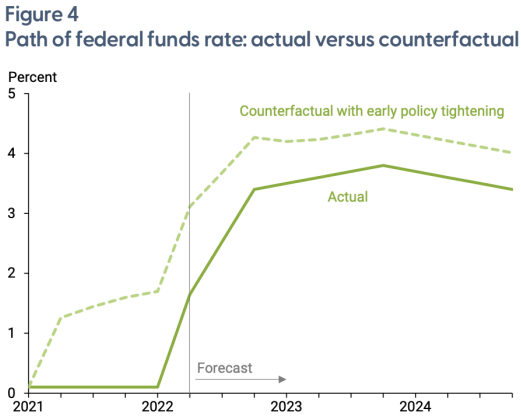


Figure shows actual and counterfactual FFR paths. Source: Barnichon (2022).

Evaluating post-2021 rate hikes

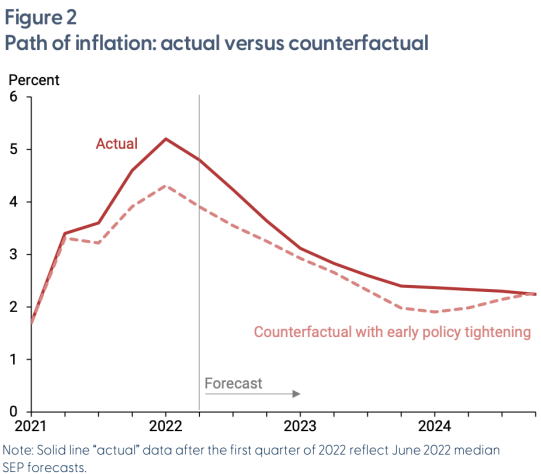


Figure shows actual and counterfactual inflation paths. Source: Barnichon (2022).

Evaluating post-2021 rate hikes

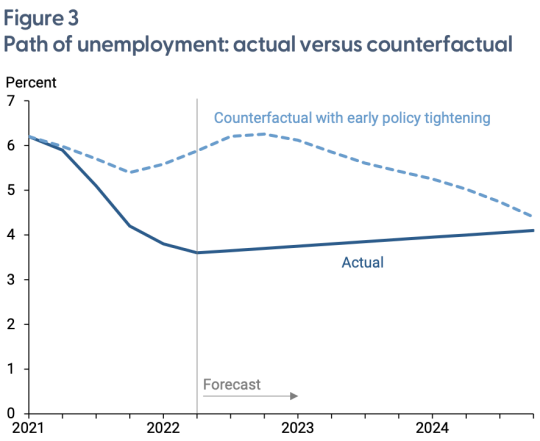
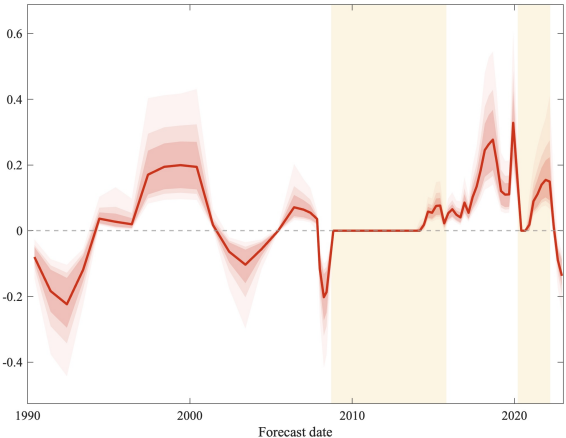


Figure shows actual and counterfactual unemployment paths. Source: Barnichon (2022).

Optimal policy decisions post-1990

Exercise: could **monetary shocks** have improved unemployment & inflation outcomes?



Source: own computations, based on Barnichon and Mesters (2023).

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Policy causal effect extrapolation

- **Model environments**

  a) **RANK**: standard medium-scale DSGE model
  Habit formation, variable capacity utilization, inv. adjustment costs, & price + wage indexation.

  b) **HANK**: HA + standard quantitative NK-DSGE frictions
  Match micro MPC evidence, replace habit formation with sticky information, otherwise = RANK.

  Extension: adding **behavioral frictions** in price- & wage-setting, à la Gabaix (2020).

- **Results**

  For detailed discussion and results see Caravello, McKay, and Wolf (2026).

  - Find: all models can match the transitory policy targets $\theta_{x,z}$ well, and equally so
  - **RANK** & **HANK** extrapolate to persistent shocks similarly, **behavioral** very different
    Implies: *all* counterfactuals will be essentially identical for RANK/HANK, but possibly different for behavioral extrapolation. HANK irrelevant for aggregate monetary policy counterfactuals!

The average business cycle

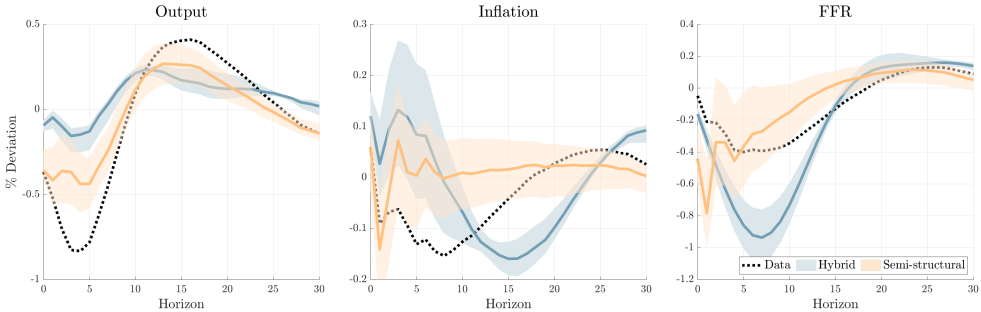
How **volatile** would macro aggregates be under flexible inflation targeting?

	Inflation	Output	Interest Rate
Actual	1.978	3.050	2.927
Counterfactual			
Hybrid			
RANK	1.674	1.795	3.748
	(1.607, 1.772)	(1.591, 2.091)	(3.407, 4.049)
HANK	1.634	1.774	3.496
	(1.558, 1.733)	(1.554, 2.101)	(3.203, 3.758)
B-RANK	1.679	1.577	3.692
	(1.582, 1.859)	(1.339, 1.831)	(3.328, 4.117)
B-HANK	1.666	1.543	3.530
	(1.594, 1.844)	(1.297, 1.811)	(3.213, 3.871)
Semi-structural	1.795	2.153	2.841
	(1.628, 1.912)	(1.910, 2.422)	(2.512, 3.185)

The average business cycle

Simple way of seeing where this is coming from: cnfct'l for "main business-cycle shock"

We will discuss this shock and its identification in greater detail in Lecture Note 10.

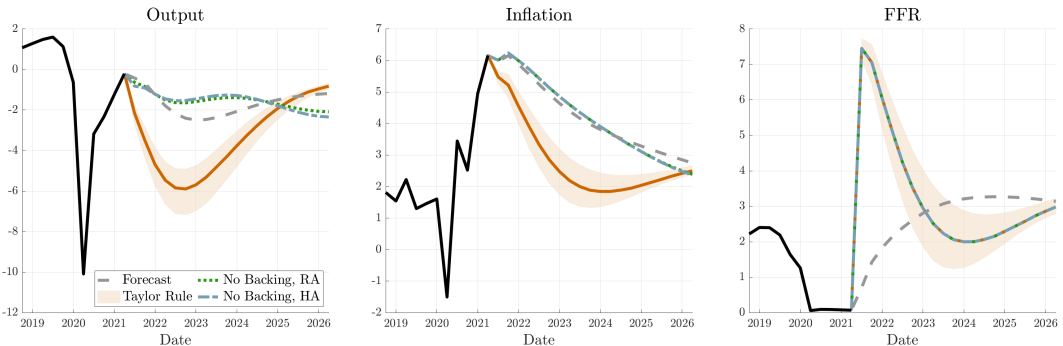


Takeaway: conclusion is immediate from two empirical moments

(i) $\exists$ **reduced-form shock** that moves y and not π + (ii) **rate cuts** boost y

Post-2021 and the policy regime

Exercise: post-2021 rate hikes under monetary and fiscal dominance eq'm selection



Source: Caravello, McKay, and Wolf (2026). Key: regime really matters, RANK vs. HANK much less.

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Summary & takeaways

- This lecture: applications of our identification results to **policy evaluation**
- Important: informs both semi-structural & fully structural work
  1. *Semi-structural*: use empirical evidence on **policy causal effects** alone
     - In practice only have a few estimated "treatments." Can use those for some counterfactuals, or to ask whether policy was sub-optimal.
     - Key task for future empirical work: get more treatments
  2. *Structural*: role of model structure is causal effect **extrapolation**
     - Sufficient statistics are useful to *explain* model results: what drives conclusions?
     - Does friction XYZ matter? only if it changes extrapolation